

Factoring Algebraic Expressions (Part II) Homework

Factor.

① $x^2 - 9$

$(x-3)(x+3)$

② $y^2 - 36$

$(y+6)(y-6)$

③ $b^2 - 100$

$(b-10)(b+10)$

④ $x^2 - 4$

$(x-2)(x+2)$

⑤ $y^2 - 25$

$(y-5)(y+5)$

⑥ $b^2 - 16$

$(b-4)(b+4)$

⑦ $x^2 - 64$

$(x-8)(x+8)$

⑧ $25y^2 - 4$

$(5y-2)(5y+2)$

⑨ $16b^2 - 49$

$(4b-7)(4b+7)$

⑩ $100x^2 - 1$

$(10x-1)(10x+1)$

⑪ $81y^2 - 16$

$(9y-4)(9y+4)$

⑫ $4 - b^2$

$(2-b)(2+b)$

⑬ $25 - x^2$

$(5+x)(5-x)$

⑭ $49 - y^2$

$(7-y)(7+y)$

⑮ $9 - 100b^2$

$(3-10b)(3+10b)$

⑯ $64 - 9x^2$

$(8-3x)(8+3x)$

⑰ $-y^2 + 64$

$(8-y)(8+y)$

⑱ $-16 + b^2$

$(b-4)(b+4)$

⑲ $-x^2 + 81$

$(9-x)(9+x)$

⑳ $-4 + y^2$

$(y-2)(y+2)$

㉑ $x^2 + 4xy + 4y^2$

$(x+2y)(x+2y)$

㉒ $a^2 + 11ab + 28b^2$

$(a+7b)(a+4b)$

㉓ $x^2 - 6xy - 16y^2$

$(x-8y)(x+2y)$

㉔ $a^2 - 2ab + b^2$

$(a-b)(a-b)$

㉕ $x^2 - 12xy + 35y^2$

$(x-5y)(x-7y)$

㉖ $2a^2 + 11ab - 6b^2$

$(2a-b)(a+6b)$

㉗ $3x^2 + 11xy + 6y^2$

$(3x+2y)(x+3y)$

㉘ $4a^2 - b^2$

$(2a-b)(2a+b)$

㉙ $x^2 - y^2$

$(x-y)(x+y)$

㉚ $25a^2 - 4b^2$

$(5a-2b)(5a+2b)$

㉛ $100x^2 - 49y^2$

$(10x-7y)(10x+7y)$

㉜ $a^2 - 16b^2$

$(a-4b)(a+4b)$